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Database Foundations

2-6

Entity Relationship Modeling (ERDs)

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Objectives

- This lesson covers the following objectives:
  - Describe data modeling
  - Explain "implementation-free" as it relates to data models and database design implementation
  - List the four goals of entity relationship modeling
  - Identify an entity relationship diagram (ERD)
  - Map relationships using ERDish
  - Construct ERD components that represent entities and attributes according to diagramming conventions

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Purpose of Conceptual Modeling

- When you design a house, you eventually want to see the house built
- Even if you don't do the actual construction, you need to understand the builder's terms to help them take your conceptual design and make it a physical reality
- The database conceptual model can be used for further discussion between designers, database administrators, and application developers

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The architect is trained in the skills of translating ideas into models. The architect listens to the description of the ideas and asks many questions that are then put into a diagram (the blueprint) that allows for discussion and analysis, giving advice, describing sensible options, documenting it, and confirming it with the future homeowners. This diagram provides the future homeowners with a plan of the home they want.

A building contractor needs the blueprint of the house with an exact description of the materials to be used, the size of the roof beams, the capacity of the plumbing, and many other specifications. The builder follows the plan and has the knowledge to construct what is in the blueprint into a physical reality.

Purpose of Conceptual Modeling

- A conceptual model is important to a business because it:
  - Describes the exact information needs of the business
  - Facilitates discussion
  - Prevents mistakes and misunderstandings
  - Forms a sound basis for physical database design
  - Documents the processes (also known as business rules) of the business
  - Takes into account regulations and laws governing this industry

Conceptual Modeling

- A conceptual model is a formal model which:
  - Describes the things of significance to an organization (entities)
  - Identifies the highest-level relationships between the different entities, but may or may not include cardinality and nullability
  - Does not specify the attributes or the unique identifier for each entity

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The conceptual modeling has a formal analysis and design method that uses a set of guidelines and rules to capture the semantics of a domain. Formal methods include textual or graphical notations to create, present, validate, and manipulate data models. It clarifies identification of entities, and relationships. It provides a basis for discussion and refinement.

Logical Modeling

- The logical data model :
  - Describes the data in as much detail as possible, without regard to how it will be physically implemented in the database
  - Is normally derived from a conceptual data model
  - Includes all entities, attributes, UIDs and relationships as well as optionality and cardinality of these items
- The logical model is illustrated using an ERD

Steps to Create a Logical Model

1. Identify entities
2. **Identify attributes (including optionality)**
3. **Identify unique identifiers**
4. **Determine relationships (including optionality and cardinality)**

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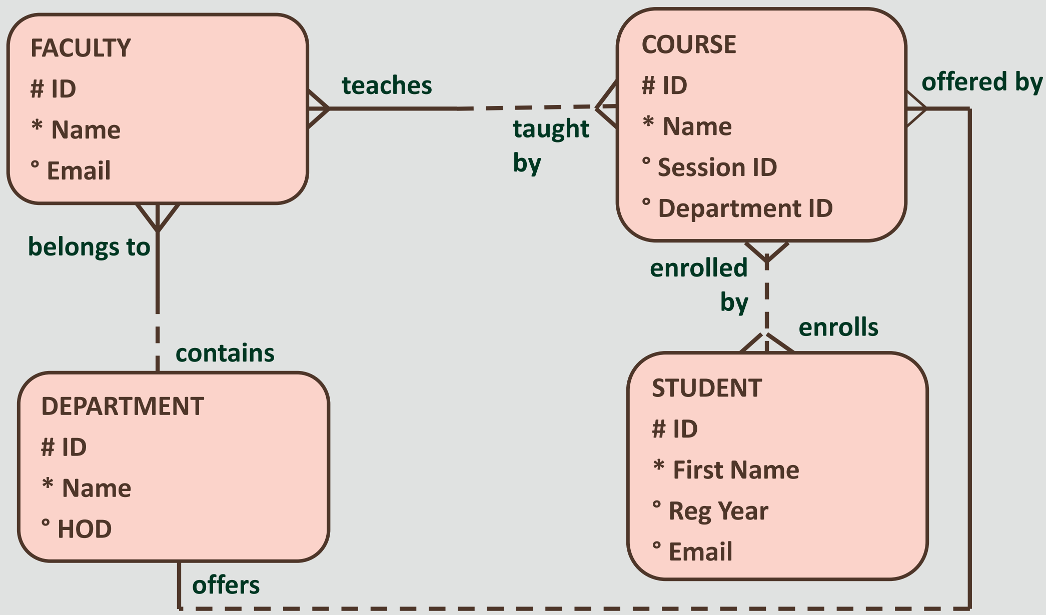
Logical Modeling: Example

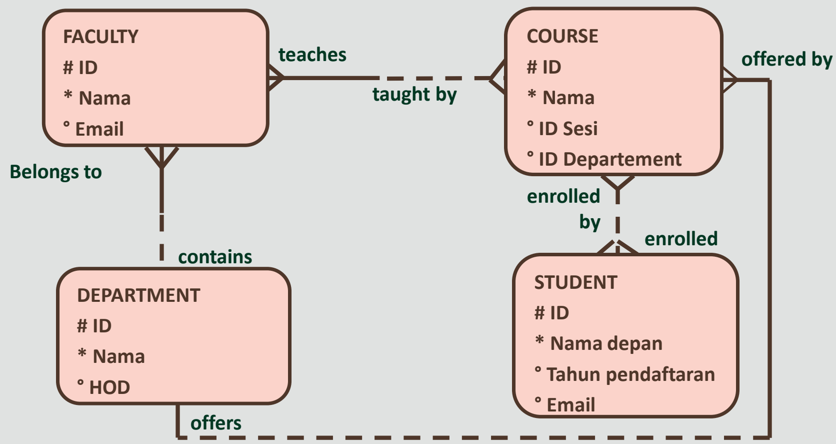


Entity Relationship Diagram (ERD)

- An ERD is a model that identifies the concepts or entities that exist in a system and the relationships between those entities
- It serves several purposes:
  - The database analyst/designer gains a better understanding of the information to be contained in the database through the process of constructing the ERD
  - It serves as a documentation tool
  - It is used to communicate the logical structure of the database to users. In particular, it effectively communicates the logic of the database to users

Entity Relationship Diagram (ERD)

- An ERD can be used to represent the data requirements of a business, regardless of the type of database that is used, and even in the absence of one
- A graphical representation of entities and their relationships to each other

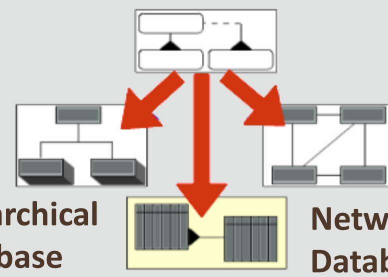


Implementation-Free Models

- A good logical data model stays the same regardless of the type of database system that is eventually built—or implemented—on
- This is what “implementation-free” model means

Database Types

ER Model



Hierarchical Database

Network Database

Relational Database

The data model should stay the same even if a database is not used at all; for example, when the data is eventually stored on pieces of paper in a filing cabinet.

Entity Relationship Model (ERM)

- Is derived from business specifications, and its intended goal is to create a clear picture of the information that will be stored in a future database
- Is a list of all entities and attributes as well as all relationships between the entities that are of importance
- Provides background information such as entity descriptions, data types, and constraints
- Does not require a diagram, but the diagram is typically a very useful tool

Goals of ER Modeling

- Capture all required information
- Ensure that information appears only once
- Model no information that is derivable from other information that is already modeled
- Locate information in a predictable, logical place

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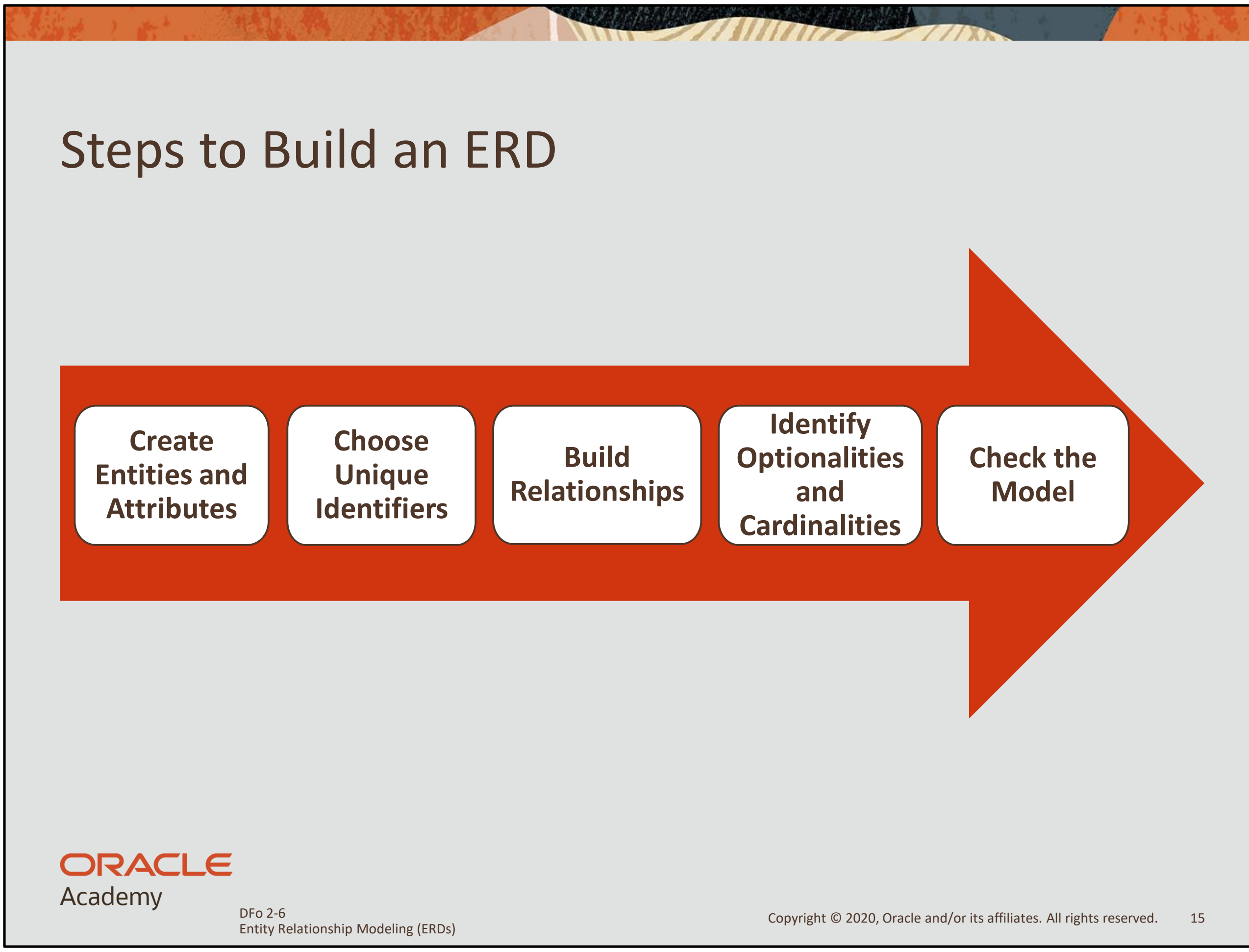
Steps to Build an ERD
Create Entities and Attributes
Choose Unique Identifiers
Build Relationships
Identify Optionalities and Cardinalities
Check the Model
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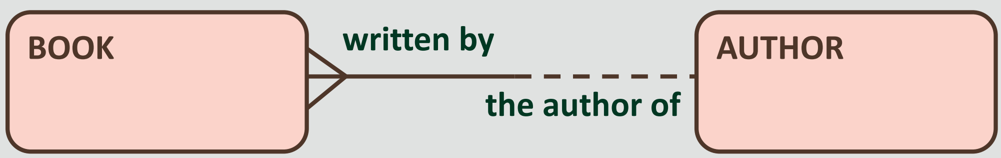
Create ERDish sentences to represent ERDs

- ERDish is the vocabulary used to clearly communicate the business rules that are captured on an ERD
- Use ERDish language to state relationships between entities in an ERD
- Simply break down each ERDish sentence into its components



Data modeling uses industry-specific terminology as well, which we will call ERDish for the purposes of this class. It will give you a common language both when collecting the business rules from your client and communicating them to the Database Administrators who will implement your design.

You have already been speaking and writing it, when you identified relationships and specified optionality and cardinality.

Components of ERDish

- EACH
- Entity A
- OPTIONALITY (must be/may be)
- RELATIONSHIP NAME
- CARDINALITY (one and only one/ one or more)
- Entity B

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ERDish Example

Because a relationship has two sides, first read one side from left to right.

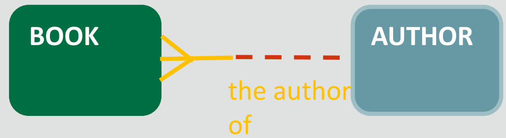
1. EACH
2. **BOOK** (entity A)
3. **MUST BE** (optionality, solid line)
4. WRITTEN BY(relationship name)
5. **ONE (AND ONLY ONE)** (cardinality, single toe)
6. AUTHOR(entity B)

Next, read the relationship from right to left.

1. EACH
2. AUTHOR (entity B)
3. **MAY BE** (optionality, dotted line)
4. **THE AUTHOR OF**(relationship name)
5. ONE OR MORE (cardinality, crow's foot)
6. **BOOK** (entity A)

Case Scenario

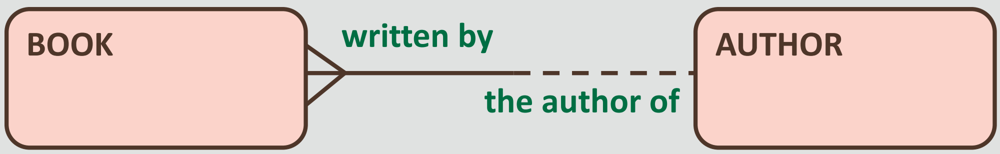
1. EACH
2. **BOOK** (entity A)
3. **MUST BE** (optionality, solid line)
4. **WRITTEN BY**(relationship name)
5. **ONE AND ONLY ONE** (cardinality, single toe)
6. **AUTHOR**(entity B)



1. EACH
2. **AUTHOR** (entity B)
3. **MAY BE** (optionality, dotted line)
4. **THE AUTHOR OF**(relationship name)
5. **ONE OR MORE** (cardinality, crow's foot)
6. **BOOK** (entity A)

Validating the Relationship

- Reexamine the ERD and validate the relationship



- Each BOOK must be written by one and only one AUTHOR
- Each AUTHOR may be the author of one or more BOOK

Sporting Goods Business Scenario

- I'm a manager of a sporting goods wholesale company that operates worldwide to fill orders of retail sporting goods stores. The stores are our customers (some of our people prefer to call them our clients)
- Right now we have fifteen customers worldwide, but we're trying to expand our customer base by about 10% each year starting this year

Sporting Goods Business Scenario

- Our two biggest customers are in the United States: Big John's Sports Emporium in San Francisco, California, and Women's Sports in Seattle, Washington
- For each customer, we must track an ID and a name. We may also track an address (including the city, state, zip code, and country) and a phone number
- We maintain warehouses in different regions to fill our customer orders

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Sporting Goods Business Scenario

- For each order, we must track an ID. We may also track the date ordered, date shipped, and payment type if the information is available. Our order entry personnel are well versed in our product line.
- We hold frequent meetings with Marketing to learn about new products. The result is better customer satisfaction because we can answer customer questions.

Sporting Goods Business Scenario

- We deal with a few select customers and maintain a specialty product line. For each product, we must know the ID and name. Occasionally we must also know the description, suggested price, and unit of sale
- When necessary we also want to be able to track very long descriptions of our products and pictures of our products

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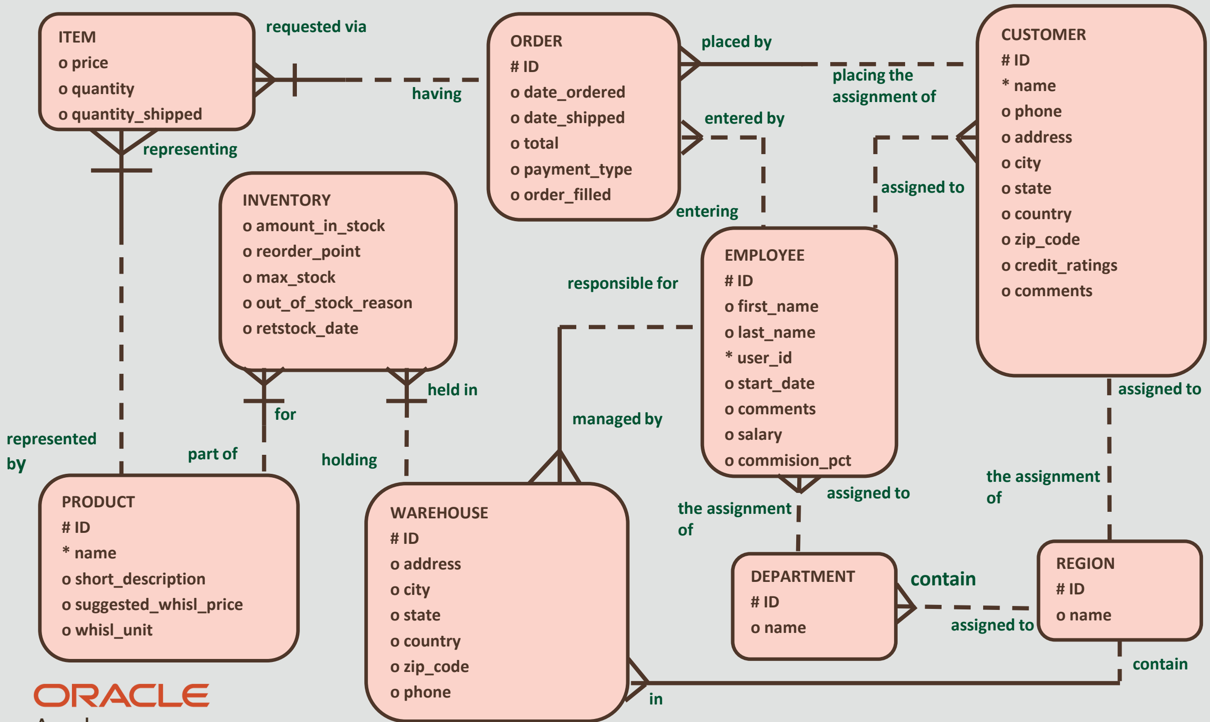
Sample Solution for Sporting Goods ERD



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Project Exercise

- DFo_2_6_Project
  - Oracle Baseball League Store Database
  - Entity Relationship Modeling

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Summary

- In this lesson, you should have learned how to:
  - Describe data modeling
  - Explain "implementation-free" as it relates to data models and database design implementation
  - List the four goals of entity relationship modeling
  - Identify an ERD
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